BT-201 – ENGINEERING PHYSICS

MODULE-WISE MOST IMPORTANT QUESTIONS (WITH YEARS)

Based strictly on PYQs: Nov-2019 → Jun-2025

MODULE–1: Wave Nature of Particles & Schrödinger Equation

1. **Derive energy eigenvalues and wave function for a particle in one-dimensional box**

   Asked in: **Nov 2019, Jun 2020, Jun 2022, Dec 2024**

2. **State and prove Heisenberg's Uncertainty Principle**

   Asked in: **Nov 2019, Dec 2020, Jun 2023, Jun 2025**

3. **Derive time-dependent and time-independent Schrödinger wave equation**

   Asked in: **Dec 2020, Jun 2022, Jun 2025**

4. **Physical significance of wave function**

   Asked in: **Jun 2022, Jun 2025**

5. **Establish relation between phase velocity and group velocity**

   Asked in: **Jun 2020, Nov 2022, Jun 2023**

MODULE–2: Wave Optics

1. **Young's Double Slit Experiment – derivation of fringe width with diagram**

   Asked in: **Nov 2019, Jun 2023, Dec 2024**

2. **Newton's Rings – formation and expression for diameter of rings**

   Asked in: **Jun 2020, Jun 2022, Dec 2023, Dec 2024**

3. **Michelson Interferometer – construction and working**

Asked in: **Jun 2020, Dec 2023, Jun 2025**

4. **Fraunhofer diffraction due to single slit**

   Asked in: **Nov 2019, Jun 2020, Dec 2024**

5. **Rayleigh criterion and resolving power of grating**

   Asked in: **Jun 2023, Jun 2025**

MODULE–3: Solid State Physics

1. **Explain Fermi level in intrinsic and extrinsic semiconductors**

   Asked in: **Dec 2023, Dec 2024**

2. **Kronig–Penney model (theory / conclusions)**

   Asked in: **Nov 2019, Jun 2023, Dec 2024**

3. **Hall Effect – derive Hall coefficient and Hall voltage**

   Asked in: **Jun 2020, Nov 2022, Dec 2024**

4. **P-N junction diode – working and V-I characteristics**

   Asked in: **Jun 2023, Nov 2022, Dec 2024**

5. **Zener diode – working and characteristics**

   Asked in: **Nov 2019, Jun 2022, Jun 2025**

MODULE–4: Lasers & Optical Fiber

1. **Construction and working of He-Ne laser with diagram**

   Asked in: **Jun 2020, Dec 2023, Nov 2022, Jun 2025**

2. **Einstein's A and B coefficients – derivation and relation**

   Asked in: **Jun 2020, Nov 2022, Jun 2025**

3. **Population inversion and stimulated emission**

   Asked in: **Jun 2020, Dec 2023**

4. **Numerical aperture and acceptance angle of optical fiber (derivation + numericals)**

   Asked in: **Dec 2020, Jun 2022, Dec 2024, Jun 2025**

5. **Applications and properties of LASER**

   Asked in: **Nov 2019, Jun 2023, Jun 2025**

MODULE–5: Electrostatics & Maxwell's Equations

1. **Derive Maxwell's equations in vacuum**

   Asked in: **Dec 2020, Jun 2022, Jun 2025**

2. **Derive continuity equation for current density**

   Asked in: **Jun 2020, Dec 2024, Jun 2025**

3. **Gauss theorem – statement and proof**

   Asked in: **Nov 2019, Dec 2020, Jun 2023**

4. **Poynting theorem and Poynting vector**

   Asked in: **Dec 2020, Dec 2024**

5. **Gradient, Divergence and Curl (definitions / numericals)**

   Asked in: **Dec 2020, Jun 2023**

HIGH-PROBABILITY MODULES (EXAM VIEW)

1. **Module-2: Wave Optics**
2. **Module-4: Lasers & Fiber Optics**
3. **Module-1: Quantum Mechanics**

FULL PREDICTED MODEL QUESTION PAPER (2025)

Time: 3 Hours

Maximum Marks: 70

Instructions

1. Attempt **any FIVE** questions.
2. All questions carry **equal marks (14 marks each)**.
3. Draw neat diagrams wherever required.
4. Assume suitable data if necessary.

Q1. (MODULE–1: Wave Nature & Quantum Mechanics)

a) Derive the **energy eigenvalues and eigen functions** for a particle confined in a **one-dimensional potential box**.

(7 marks)

b) State and explain the **physical significance of wave function**.

(7 marks)

Q2. (MODULE–1: Quantum Mechanics)

a) State and derive **Heisenberg's Uncertainty Principle**.

(7 marks)

b) Distinguish between **phase velocity and group velocity**.

Derive the relation between them.

(7 marks)

Q3. (MODULE–2: Wave Optics)

a) Explain **Young's Double Slit Experiment** and derive the expression for **fringe width**.

(7 marks)

b) Explain the formation of **Newton's Rings** and obtain the expression for the **diameter of nth dark ring**.

(7 marks)

Q4. (MODULE–2: Interference & Diffraction)

a) Describe the **construction and working of Michelson Interferometer**.

(7 marks)

b) Explain **Fraunhofer diffraction due to a single slit** with necessary theory.

(7 marks)

Q5. (MODULE–3: Solid State Physics)

a) Explain **Fermi level** in **intrinsic and extrinsic semiconductors** with energy band diagrams.

(7 marks)

b) Explain the **Kronig–Penney model** and discuss the **origin of energy bands**.

(7 marks)

Q6. (MODULE–3: Semiconductor Devices)

a) Explain the **working of P–N junction diode** with **V–I characteristics**.

(7 marks)

b) Explain **Zener diode** and its application as a **voltage regulator**.

(7 marks)

Q7. (MODULE–4: Lasers & Optical Fiber)

a) Derive the relation between **Einstein's A and B coefficients**.

Explain the concept of **population inversion**.

(7 marks)

b) Explain the **construction and working of He–Ne laser** with diagram.

(7 marks)

Q8. (MODULE–5: Electrostatics & Maxwell's Equations)

a) State and derive **Maxwell's equations in free space**.

(7 marks)

b) Explain **Poynting theorem** and define **Poynting vector**.

(7 marks)

Module Coverage Check

- Module-1 → Q1, Q2
- Module-2 → Q3, Q4
- Module-3 → Q5, Q6
- Module-4 → Q7
- Module-5 → Q8

Paper pattern matches **RGPV style** and is based on **PYQs (2019–2025)**.
